Expt No: 7 Flow control at Data link
Date: 8/8/24 layer using Sliding Window Protocol.

AIM:-

To write a Program to implement flow control at the Data link layer using the sliding window protocol and to simulate the flow of frames from one node to another

ALGORITHM:-

1. Start the program and enter the total number of frames, window size and the frame number to be simulated as lost.
2. Initialize the sender window and sequence numbers
3. Send frames within the current window from node A to node B.
4. If a frame is lost the receiver discards subsequent out of order frames and the sender retransmits from the lost frame using GoBack - N
5. If there is no loss the receiver sends acknowledgements and the sender slides the window forward.
6. Repeat until all frames are successfully transmitted and acknowledged.

PROGRAM 2

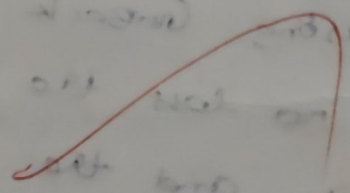
```
def simulate(total_frames, window_size, lost_frame):
    base = 0
    next_seq = 0
    while base < total_frames:
    window_end = min(base + window_size - 1, total_frames
    print("\n--------")
    print("Sender window : [Frame{base} -
                            Frame{window_end}]")

    while next_seq < base + window_size and
          next_seq < total_frames:
        print(f"Node A => Sending Frame {next_seq}")
        if next_seq == lost_frame:
            print(f" [Frame {next_seq} Lost in
                    transit]")
            next_seq += 1
    if lost_frame is not None and base <= lost_frame
                                < next_seq:
        print(f"Node B --> Frame {lost_frame}
            missing. Discarding out of order frames
                                after it")
        print(f"Node A --> Timeout! Resending window
            starting from Frame {lost_frame}.")
        next_seq = lost_frame
        lost_frame = None
        continue
    for i in range(base, next_seq):
        print(f"Node B --> Received frame {i},
                    sending Ack {i}")
        print(f"Node A --> Ack {next_seq - 1} received
                Sliding window forward")
        base = next_seq
    print("\n--------")
```

```
print(f" All {total_frames} frames successfully
        transmitted from Node A to Node B")

def main():
    print("\n== SLIDING WINDOW PROTOCOL-FLOW CONTROL
            SIMULATION (GoBack N) == \n")
    total_frames = int(input("Enter total number of
                    frames to send : "))
    window_size = int(input("Enter sender's
                    window size : "))
    lost_frame = int(input(f" Enter frame number
        to simulate as LOST \n(0 to {total_frames-1}
                    Enter -1 for no loss): "))
    if lost_frame == -1:
        lost_frame = None
    simulate(total_frames, window_size, lost_frame)
if __name__ == "__main__":
    main()
```

(2)



OUTPUT

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=== SLIDING WINDOW PROTOCOL - FLOW CONTROL SIMULATION (Go-Back-N) ===

Enter total number of frames to send   : 8
Enter sender's window size             : 4
Enter frame number to simulate as LOST
(0 to 7, enter -1 for no loss)         : 5

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Sender Window : [Frame 0 - Frame 3]
Node A -->Sending Frame 0
Node A -->Sending Frame 1
Node A -->Sending Frame 2
Node A -->Sending Frame 3
Node B -->Received Frame 0, sending ACK 0
Node B -->Received Frame 1, sending ACK 1
Node B -->Received Frame 2, sending ACK 2
Node B -->Received Frame 3, sending ACK 3
Node A -->ACK 3 received. Sliding window forward.

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Sender Window : [Frame 4 - Frame 7]
Node A -->Sending Frame 4
Node A -->Sending Frame 5
      (Frame 5 LOST in transit!)
Node A -->Sending Frame 6
Node A -->Sending Frame 7
Node B -->Frame 5 missing. Discarding out-of-order frames after it.
Node A -->Timeout! Resending window starting from Frame 5.

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Sender Window : [Frame 4 - Frame 7]
Node A -->Sending Frame 5
Node A -->Sending Frame 6
Node A -->Sending Frame 7
Node B -->Received Frame 4, sending ACK 4
Node B -->Received Frame 5, sending ACK 5
Node B -->Received Frame 6, sending ACK 6

Node B -->Received Frame 7, sending ACK 7
Node A -->ACK 7 received. Sliding window forward.

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All 8 frames successfully transmitted from Node A to Node B.
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Result :-

The program to implement flow control at the Data link layer using sliding window protocol was successfully written in Python, executed and verified by stimulating the flow of frames between two nodes.